

Technical University of Munich | Practical course, extended to a paper | Supervisor: M. Sabanayagam

Invariance vs Robustness: What Orders a Model’s Adversarial Robustness

A threat-matched margin-to-Lipschitz account, from small classifiers to frozen foundation-model encoders

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Current status: a combined paper at roughly 23 pages with a central concept figure and five empirical regimes, plus a full proofs-and-details appendix; code and result files reproduce every number. A pitch deck and an internal reader script accompany this summary. This is a two-page summary for feedback, not a submission plan.

Summary. Whether shift-invariance helps or hurts adversarial robustness is contested: Ge et al. (2021) prove a fully shift-invariant classifier can be far less robust than a fully connected one, while anti-aliasing and equivariance work (Zhang 2019; Saha and Gokhale 2024; Wang 2025) reports the opposite correlation. Every “helps” result, though, used a gradient attack (FGSM or PGD), never a parameter-free strong attack. We organise both sides around one quantity. A shift-invariant linear classifier separates classes only through their projection onto the invariant subspace, with an exact margin identity (the distance between the projected convex hulls, Proposition 1). The surviving robustness is summarised by a threat-matched, scale-free margin-to-Lipschitz ratio η/L , a Lipschitz–margin certificate (Hein 2017; Tsuzuku 2018) that we complete to a *two-sided bracket* $\eta/L \leq r \leq \eta/\alpha$ with condition number $\kappa = L/\alpha \in [1.6, 2.4]$ (Proposition 2). Across capacity-matched dissections and, importantly, within single frozen CLIP encoders where training cannot confound, η/L orders the adversarial robust radius while shift-consistency does not; on modern encoders shift-consistency is saturated (0.963–0.988) and cannot rank at all. The “invariance helps” correlation is a *weak-attack artifact*: it appears under FGSM and vanishes under AutoAttack. To rank already-robust models, where η/L no longer separates, the *shape* of the input gradient (an ℓ_1/ℓ_2 anisotropy) tracks robustness across thirty RobustBench models (Spearman -0.79), through a proposed spread-over-dimensions deficit.

1 Problem and motivation

Image classifiers are often called shift-invariant when circular translations of the input do not change the prediction. The published evidence on what this does to adversarial robustness is mixed. Ge et al. (2021) show a fully shift-invariant network can be fooled by an ℓ_2 perturbation of order $1/\sqrt{d}$, where a fully connected one needs order 1. Anti-aliased pooling and equivariance work (Zhang 2019; Saha and Gokhale 2024; Grabinski 2023; Wang 2025) reports that operators which increase shift-invariance also increase robustness. A separate line proves a trade-off between geometric and adversarial robustness (Kamath et al. 2021; Tramèr et al. 2020). The practitioner is left without a rule for when invariance is safe.

Two under-tested assumptions. The “helps” line only ever measured robustness with gradient attacks (FGSM/PGD), which over-report robustness, and it never tested whether the shift-consistency *metric* actually orders robustness across models. Both gaps are testable.

The gap. There was no account that (i) says what discriminative signal survives a given invariance, (ii) turns that into a robustness certificate with a converse, and (iii) is validated against a standardized strong attack at matched capacity and, ideally, inside a single frozen model. This project supplies all three.

2 The quantity and its theory

Proposition 1 (invariant margin). For a finite group acting orthogonally, an invariant linear classifier sees the data only through the group-average projector P_G ; its maximum ℓ_2 margin is $\frac{1}{2} \text{dist}(\text{conv } P_G X_+, \text{conv } P_G X_-)$, the separation of the class averages. The cyclic-shift case is the DC projection and recovers Ge’s $2/\sqrt{d}$ collapse, now for any finite separable dataset, not one image.

The criterion. The surviving robustness is summarised by η/L : the mean margin η over the mean dual-norm gradient L on clean, correctly classified images. We *threat-match* the norm to the attack ($q=1$ for ℓ_∞ , $q=2$ for ℓ_2 ; the mismatched norm gives the wrong sign), and note it is *scale-free*, so robustness attaches to the ratio, not to margin or Lipschitz alone. No attack is needed to compute it.

Proposition 2 (two-sided bracket). With an L -Lipschitz and α -co-Lipschitz condition toward the boundary, $\eta/L \leq r \leq \eta/\alpha$, so the true radius is trapped, not merely lower-bounded; the gap is $\kappa = L/\alpha$, exact ($\kappa=1$) for linear invariant features and $\kappa \in [1.6, 2.4]$ empirically. This is the converse the one-sided Lipschitz–margin certificate lacked.

Further theory (appendix). A convolution-square-pool layer computes power-spectrum coordinates; a degree-three bispectrum invariant separates the phase-coded classes the power spectrum cannot. On orbit-closed data invariance is *margin-free* (invariant and unconstrained max-margin values coincide); it helps only on near-orthogonal cluster geometry, where a trajectory-level selection theorem is the open lazy-regime question.

3 Main empirical evidence

η/L orders robustness; shift-consistency does not. Across settings, the threat-matched ratio tracks the robust radius or AutoAttack robust accuracy, while shift-consistency is uncorrelated or anti-correlated.

setting (attack / metric)	η/L	consist.
CIFAR-10 standard (min-norm r)	+0.998	−0.33 n.s.
PreActResNet-18 AT (AA ℓ_∞)	+0.91	−0.81
ImageNet-100 AT (AA ℓ_∞)	+0.90	−0.87
16 CLIP encoders, per image (r)	+0.74 to +0.96	≈ 0

Pearson or Spearman as appropriate. The per-image rows are *within* a single frozen encoder, so training is not the confound.

The “helps” effect is a weak-attack artifact.

Re-running the “helps” regime (standard training, FGSM/PGD) on our operators and on Saha (2024) and Wang (2025), the ordering appears under FGSM and disappears under AutoAttack, which drops every non-robust model to near zero; on frozen CLIP encoders FGSM reports 16–61% robust accuracy that AutoAttack erases.

Saturation and the per-image test. On sixteen frozen CLIP encoders shift-consistency sits at 0.963–0.988 regardless of robustness, so it cannot rank them. Image by image inside one robust encoder, η/L orders each image’s radius (Spearman +0.74 to +0.96) and per-image consistency does not.

Ranking robust models by gradient shape. Among already-robust models η/L no longer separates. The anisotropy $A = \|\nabla M\|_1 / \|\nabla M\|_2 \in [1, \sqrt{d}]$ does: over thirty RobustBench CIFAR-10 models, lower A is more robust (Spearman −0.79; −0.77 controlling for clean accuracy), replicated on an MNIST/Fashion training-strength sweep (−0.61 to −0.89).

4 Mechanism diagnostics

The margin/Lipschitz decomposition separates two mechanisms: anti-aliasing leaves the margin essentially unchanged and lowers the Lipschitz constant ($\Delta \log L \approx -0.2$), buying robustness through reduced sensitivity; exact cyclic invariance shrinks the margin ($\Delta \log \eta \approx -0.3$ to -0.4) and loses robustness, the real-data Ge collapse. For the anisotropy law we propose a spread-over-dimensions deficit $r \approx (\eta/L_1)/(1 + CA)$, $C \approx 2.65$, verified on a controlled model and thirty real ones. It is a proposed mechanism, not a network-level theorem.

5 What is already done

- **Structural theory**, audited and re-verified: the invariant margin identity (Prop. 1), the two-sided bracket (Prop. 2) with a measured κ , the power-spectrum and bispectrum span lemmas, and the orbit-rank and orbit-flip bounds.
- **Capacity-matched dissection** on MNIST, Fashion-MNIST and CIFAR-10, robust radius validated against AutoAttack (mean gap 0.014), ruling out gradient masking.
- **Adversarial-training arm** (PGD-AT, AutoAttack)

with the threat-matched η/L law and the shift-consistency anti-prediction.

- **Foundation-model regime:** sixteen frozen CLIP encoders, the saturation finding, and the per-image dissociation where training cannot confound.
 - **Anisotropy study** on thirty RobustBench models with a clean-accuracy control, plus the effective-dimension deficit.
- ## 6 Limitations and future work
- **Not a training target.** η/L is scale-free, so its direct maximizer is the constant classifier: a diagnostic, not a loss.
 - **The shape law needs range.** On low-dimensional data anisotropy is compressed and does not rank until a training-strength sweep widens the range.
 - **The deficit is proposed, not proven**, and the cluster-geometry selection theorem (lazy regime) is still open.
 - **Scale and norms.** Beyond ℓ_∞ ; larger backbones; a genuine text-adversarial axis (current CLIP paraphrase worst case is too mild).

7 Project context

This grew out of a TU Munich practical course that reproduced and extended Ge et al. (2021). The supervisor’s framing asked for a comparison against Ge et al. (2021), the two Kamath et al. trade-off papers, and Galloway et al. on batch normalization; the dissection reproduces the Galloway effect under fixed- ϵ accuracy and shows it is threshold-dependent under the radius metric. Repository: [adversarial-robustness-shift-invariance/](https://github.com/robustness-shift-invariance). Compute to date is a modest number of GPU-hours on two RTX A6000.

Selected references. Ge, Singla, Basri, Jacobs, Shift Invariance Can Reduce Adversarial Robustness, NeurIPS 2021. Kamath, Deshpande, Subrahmanyam, Balasubramanian, Can We Have It All?, NeurIPS 2021. Tramèr, Behrmann, Carlini, Papernot, Jacobsen, Fundamental Tradeoffs Between Invariance and Sensitivity, ICML 2020. Saha, Gokhale, On shift-invariance and adversarial robustness (TIPS), WACV 2024. Wang et al., Equivariance and adversarial robustness, 2025. Galloway et al., Batch Normalization Is a Cause of Adversarial Vulnerability, 2019. Zhang, Making Convolutional Networks Shift-Invariant Again, ICML 2019. Hein, Andriushchenko, Formal Guarantees on the Robustness of a Classifier, NeurIPS 2017. Tsuzuku, Sato, Sugiyama, Lipschitz-Margin Training, NeurIPS 2018. Croce, Hein, Reliable Evaluation with AutoAttack, ICML 2020. Rony et al., Decoupling Direction and Norm for Min-Norm Attacks, CVPR 2019. Weng et al., Evaluating the Robustness of Neural Networks: CLEVER, ICLR 2018.